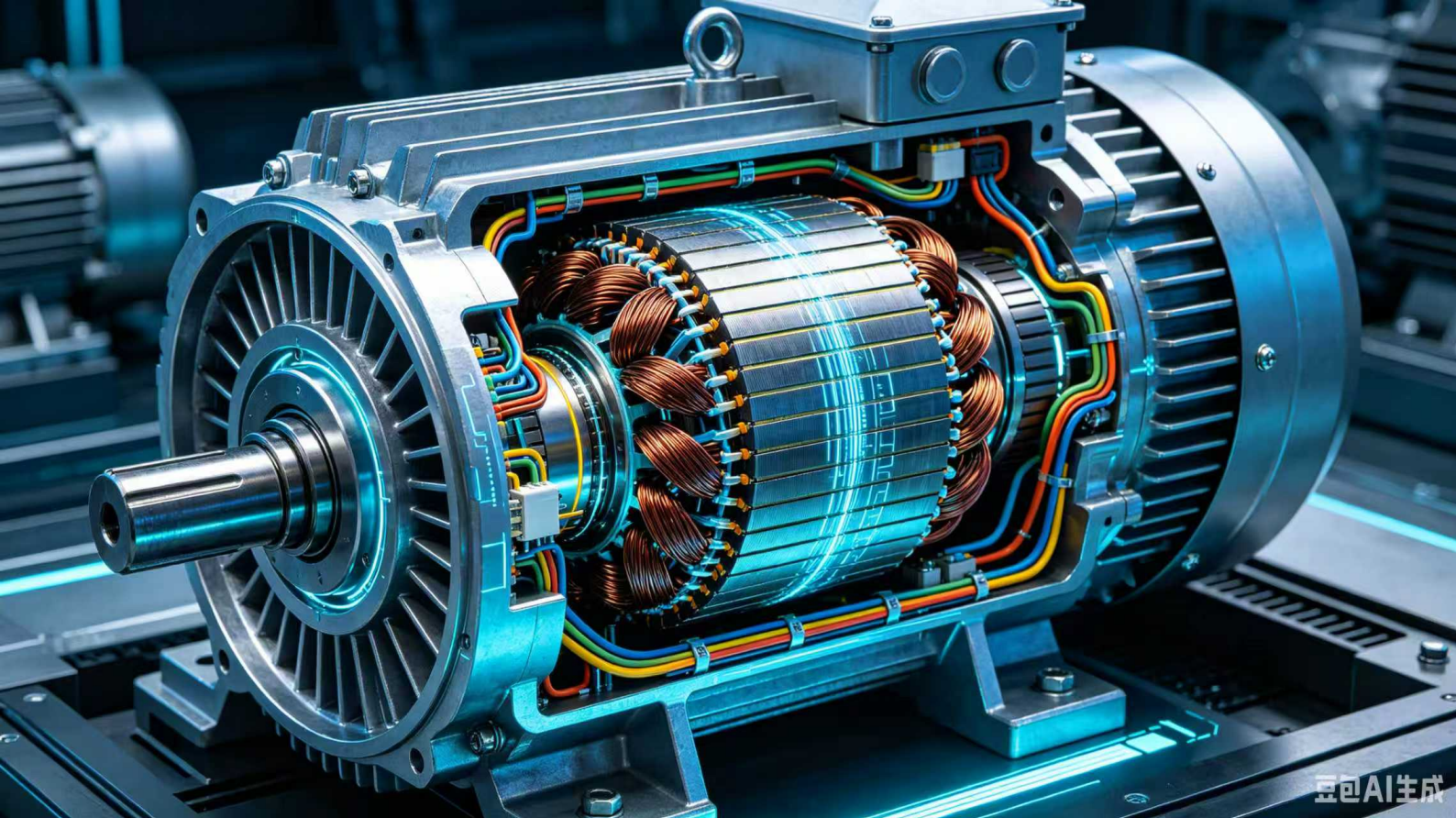




4. Triangle of Electrostatics*

Mathematics is the queen of the sciences.

– Carl Friedrich Gauss, as reported by W. Sartorius von Waltershausen (1856)

We shall encounter a number of vector identities, vector operators, and vector integrals. It would be nice if we could familiarize ourselves with them. In addition, we shall derive the “differential form” of Gauss’ Law.

4.1 Some Mathematics

4.1.1 Gradient

$\nabla = \hat{x} \frac{\partial}{\partial x} + \hat{y} \frac{\partial}{\partial y} + \hat{z} \frac{\partial}{\partial z}$ is a special vector operator. It can be applied to scalar functions to form vector functions: applied to the scalar V of the last chapter, it gives the vector $-\mathbf{E}$, as we shall see below. For a scalar spatial function $\phi(x, y, z)$, $\nabla \phi = \hat{x} \partial \phi / \partial x + \hat{y} \partial \phi / \partial y + \hat{z} \partial \phi / \partial z$ is called the **gradient** of ϕ . It is fairly straightforward to show that:

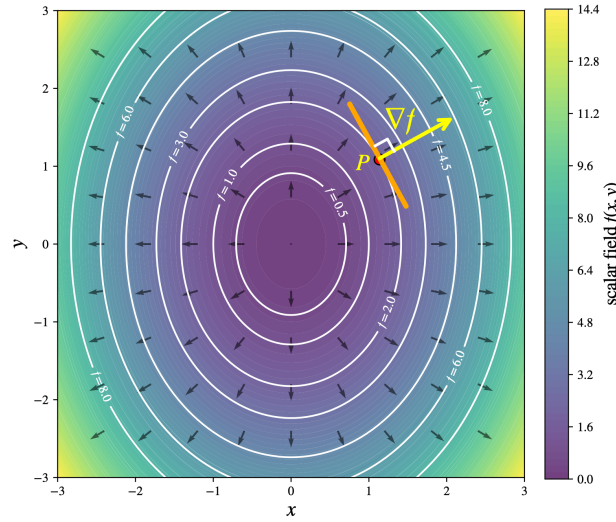


Figure 4.1: Gradient of a scalar field $\phi(x, y)$, labelled f in the figure. Contour lines are curves of constant ϕ , and the vector $\nabla\phi$ is perpendicular to the contour through that point, pointing in the direction of steepest ascent. Dark gray arrows show the direction of $\nabla\phi$.

$$\begin{aligned}
 \int_{\mathbf{r}_1}^{\mathbf{r}_2} \nabla\phi \cdot d\mathbf{r} &= \int_{\mathbf{r}_1}^{\mathbf{r}_2} \left[\hat{x} \frac{\partial\phi}{\partial x} + \hat{y} \frac{\partial\phi}{\partial y} + \hat{z} \frac{\partial\phi}{\partial z} \right] \cdot [\hat{x}dx + \hat{y}dy + \hat{z}dz] \\
 &= \int_{\mathbf{r}_1}^{\mathbf{r}_2} \left[\frac{\partial\phi}{\partial x} dx + \frac{\partial\phi}{\partial y} dy + \frac{\partial\phi}{\partial z} dz \right] \\
 &= \int_{\mathbf{r}_1}^{\mathbf{r}_2} d\phi = \Delta\phi
 \end{aligned}$$

So the line integral of $\nabla\phi$ between two points gives the change of ϕ between them, and is independent of the path taken. Now take ϕ to be the electric potential V . The definition of the potential difference gives $V(\mathbf{r}') - V(\mathbf{r}) = -\int_{\mathbf{r}}^{\mathbf{r}'} \mathbf{E} \cdot d\mathbf{l}$ along any path, and since this must agree with the line integral of ∇V for every path, we have

$$\mathbf{E} = -\nabla V \quad (4.1)$$

Example 4.1: Electric field of a point charge from its potential.

The electric potential due to a point charge q at the origin, with $V = 0$ at infinity, Eq. (3.2), is

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \frac{q}{r}$$

Using $\mathbf{E} = -\nabla V$, and since $r = \sqrt{x^2 + y^2 + z^2}$ gives $\partial r / \partial x = x/r$, and likewise for y and z , so that $\nabla r = \mathbf{r}/r = \hat{\mathbf{r}}$ and $\nabla(1/r) = -\hat{\mathbf{r}}/r^2$, we have:

$$\begin{aligned} \mathbf{E} &= -\frac{1}{4\pi\epsilon_0} \nabla \left(\frac{q}{r} \right) \\ &= \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} (\nabla r) \\ &= \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{\mathbf{r}} \end{aligned}$$

The above is extremely simple; we shall try a slightly more complicated one.

Example 4.2: Electric field of a point dipole from its potential.

The electric potential of a point dipole $\mathbf{p}$ at the origin, which Chapter 3, Eq. (3.4), obtained as the leading term for two charges $\pm q$ a small distance d apart, is

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \frac{\mathbf{p} \cdot \mathbf{r}}{r^3}$$

Using $\mathbf{E} = -\nabla V$, together with $\nabla(\mathbf{p} \cdot \mathbf{r}) = \mathbf{p}$ for a constant $\mathbf{p}$, the relation $\nabla r^3 = 3r^2 \nabla r = 3r^2 \hat{\mathbf{r}}$ that follows from the chain rule and $\nabla r = \hat{\mathbf{r}}$ of Example 4.1, and the product rule (a) of Section 4.1.3, we have:

$$\begin{aligned} \mathbf{E} &= -\frac{1}{4\pi\epsilon_0} \nabla \left(\frac{\mathbf{p} \cdot \mathbf{r}}{r^3} \right) \\ &= -\frac{1}{4\pi\epsilon_0} \frac{r^3 \nabla(\mathbf{p} \cdot \mathbf{r}) - (\mathbf{p} \cdot \mathbf{r}) \nabla r^3}{r^6} \\ &= -\frac{1}{4\pi\epsilon_0} \frac{r^3 \mathbf{p} - 3r^2 (\mathbf{p} \cdot \mathbf{r}) \hat{\mathbf{r}}}{r^6} \\ &= -\frac{1}{4\pi\epsilon_0} \frac{\mathbf{p} - 3(\mathbf{p} \cdot \hat{\mathbf{r}}) \hat{\mathbf{r}}}{r^3} \\ &= \frac{1}{4\pi\epsilon_0} \frac{3(\mathbf{p} \cdot \hat{\mathbf{r}}) \hat{\mathbf{r}} - \mathbf{p}}{r^3} \end{aligned}$$

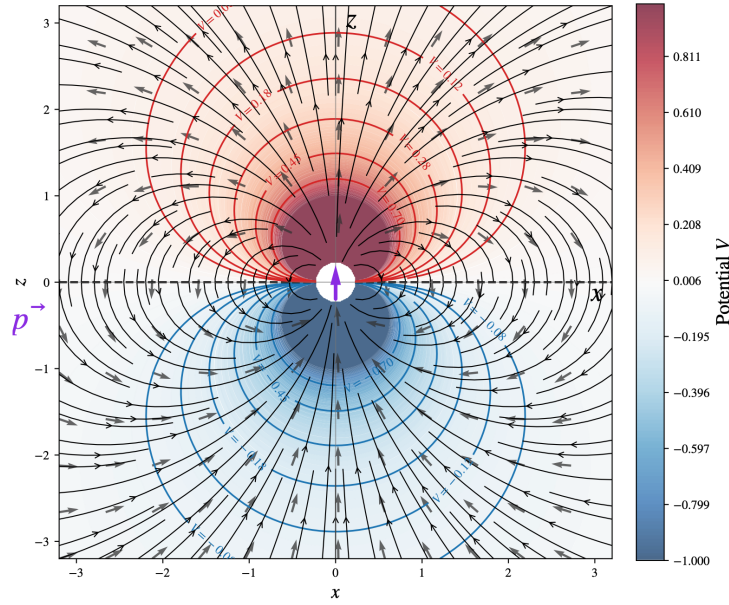


Figure 4.2: Field lines (black) and equipotentials (red for $V > 0$, blue for $V < 0$) of a point dipole $\mathbf{p}$ at the origin pointing along $+z$ (purple arrow), in the xz plane, obtained from $\mathbf{E} = -\nabla V$. Gray arrows show the direction of $\mathbf{E}$. The dashed line $z = 0$ is the equipotential $V = 0$.

Since $d\phi = \nabla\phi \cdot d\mathbf{r}$, ϕ does not change along a displacement perpendicular to $\nabla\phi$, so $\nabla\phi$ is perpendicular to the surfaces of constant ϕ ; and among displacements of a given small length, ϕ increases most along $\nabla\phi$, wherever $\nabla\phi \neq 0$. The gradient of a scalar function ϕ therefore points in the direction in which ϕ increases fastest. Hence it is very useful for finding maxima or minima, where $\nabla\phi = 0$, although a point where $\nabla\phi = 0$ can also be a saddle point. For example, when minimizing a function in scientific computing, a large family of algorithms is directly associated with it, including *conjugate gradient* and *steepest descent*. The idea is to start from a random position (or a carefully chosen initial point) and move the point *against* the gradient, until the gradient vanishes. Vanishing is necessary but not sufficient: the same condition holds at a maximum or at a saddle point, which is why the family of algorithms is larger than it first appears. In general, these problems can be high-dimensional (N -D), and the gradient of a function $f(x_1, x_2, \dots, x_N)$ in N dimensions is defined as:

$$\nabla f = \left(\frac{\partial f}{\partial x_1}, \frac{\partial f}{\partial x_2}, \dots, \frac{\partial f}{\partial x_N} \right)$$

4.1.2 Curl and Divergence

Applied to a vector function, the vector operator ∇ can produce either $\nabla \cdot \mathbf{A}$ or $\nabla \times \mathbf{A}$. Evidently, the former is a scalar function, called the **divergence** of $\mathbf{A}$; whereas the latter is a vector function, called the **curl** of $\mathbf{A}$. The actual expressions are easy to memorize: just keep in mind that both ∇ and $\mathbf{A}$ are vectors with three components, and that the derivative acts on each component of $\mathbf{A}$. So

$$\nabla \cdot \mathbf{A} = \left[\hat{x} \frac{\partial}{\partial x} + \hat{y} \frac{\partial}{\partial y} + \hat{z} \frac{\partial}{\partial z} \right] \cdot [\hat{x} A_x + \hat{y} A_y + \hat{z} A_z] = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$$

and

$$\nabla \times \mathbf{A} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix} = \hat{x} \left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right) + \hat{y} \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \right) + \hat{z} \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right)$$

The curl of a gradient is always zero, or $\nabla \times \nabla \phi \equiv 0$. The proof is straightforward. You just put in the expression of $\nabla \times \mathbf{A}$ with $A_x = \frac{\partial \phi}{\partial x}$, $A_y = \frac{\partial \phi}{\partial y}$ and $A_z = \frac{\partial \phi}{\partial z}$, and use $\partial^2 \phi / \partial y \partial z = \partial^2 \phi / \partial z \partial y$ and its two companions; each of the three components then vanishes, since the two terms in it cancel. These equalities hold, and with them the identity, whenever ϕ has continuous second derivatives.

Since $\mathbf{E} = -\nabla V$, the curl of a static electric field is zero: $\nabla \times \mathbf{E} = 0$. In the language of field theory, it means that an electrostatic field is irrotational.

4.1.3 Vector Identities

It is not difficult to derive the following equations, using the definition of ∇ :

- (a) $\nabla(fg) = (\nabla f)g + f(\nabla g)$
- (b) $\nabla \cdot (f\mathbf{v}) = (\nabla f) \cdot \mathbf{v} + f(\nabla \cdot \mathbf{v})$
- (c) $\nabla \times (f\mathbf{v}) = (\nabla f) \times \mathbf{v} + f(\nabla \times \mathbf{v})$
- (d) $\nabla \cdot (\mathbf{v} \times \mathbf{w}) = \mathbf{w} \cdot (\nabla \times \mathbf{v}) - \mathbf{v} \cdot (\nabla \times \mathbf{w})$
- (e) $\nabla \times (\mathbf{v} \times \mathbf{w}) = \mathbf{v}(\nabla \cdot \mathbf{w}) - \mathbf{w}(\nabla \cdot \mathbf{v}) + (\mathbf{w} \cdot \nabla)\mathbf{v} - (\mathbf{v} \cdot \nabla)\mathbf{w}$

4.2 Differential Form of Gauss' Law

In Chapter 2 we obtained Gauss' law in integral form, Eq. (2.1). We now examine its differential counterpart, $\nabla \cdot \mathbf{E}$. Let's consider an electrostatic field $\mathbf{E}(\mathbf{r})$ and its generating charge distribution $\rho(\mathbf{r})$. We slice a small piece of volume at a specific point $\mathbf{r}$; it is surrounded by six small rectangular facets. The volume element contains approximately $\rho(\mathbf{r})\Delta x\Delta y\Delta z$ charge.

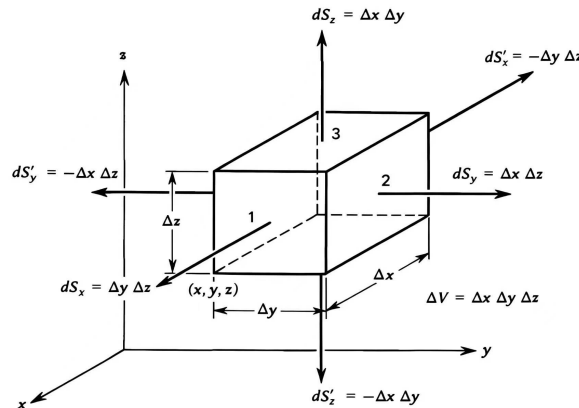


Figure 4.3: An infinitesimal volume element of sides Δx , Δy and Δz at position $\mathbf{r}$. The electric flux through each pair of opposite faces determines the net flux $\Phi_{\text{tot}} = \Phi_x + \Phi_y + \Phi_z$, which by Gauss' law equals $q_{\text{enc}}/\epsilon_0 = \rho \Delta x \Delta y \Delta z / \epsilon_0$, leading to the differential form $\nabla \cdot \mathbf{E} = \rho / \epsilon_0$.

To leading order, the electric flux through the two surfaces perpendicular to the x axis is

$$\begin{aligned}
\Phi_x &\approx \mathbf{E}\left(x + \frac{\Delta x}{2}, y, z\right) \cdot \hat{x} \Delta y \Delta z + \mathbf{E}\left(x - \frac{\Delta x}{2}, y, z\right) \cdot (-\hat{x}) \Delta y \Delta z \\
&= \left[E_x\left(x + \frac{\Delta x}{2}, y, z\right) - E_x\left(x - \frac{\Delta x}{2}, y, z\right) \right] \Delta y \Delta z
\end{aligned}$$

Each of these two terms takes the field at the center of its face times the area of the face; the exact flux is an integral of the normal component over the face, and the two agree to leading order. When Δx is small, $E_x\left(x + \frac{\Delta x}{2}, y, z\right) - E_x\left(x - \frac{\Delta x}{2}, y, z\right) \approx \frac{\partial E_x}{\partial x} \Delta x$, and we thus have $\Phi_x \approx \frac{\partial E_x}{\partial x} \Delta x \Delta y \Delta z$.

Likewise, $\Phi_y \approx \frac{\partial E_y}{\partial y} \Delta x \Delta y \Delta z$ and $\Phi_z \approx \frac{\partial E_z}{\partial z} \Delta x \Delta y \Delta z$ for the other two pairs of faces. Combining the three and applying Gauss' law, we have:

$$\left(\frac{\partial E_x}{\partial x} + \frac{\partial E_y}{\partial y} + \frac{\partial E_z}{\partial z} \right) \Delta x \Delta y \Delta z \approx \Phi_{\text{tot}} = \frac{q_{\text{enc}}}{\epsilon_0} \approx \frac{\rho(\mathbf{r}) \Delta x \Delta y \Delta z}{\epsilon_0}$$

Dividing by $\Delta x \Delta y \Delta z$ and letting the box shrink to the point $\mathbf{r}$, these approximations become exact. Noticing that the expression inside the left bracket is just $\nabla \cdot \mathbf{E}$, we have:

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0} \quad (4.2)$$

This is called the “differential form” of Gauss' law. Mathematically, Gauss' Theorem (the divergence theorem) states: for a vector field $\mathbf{A}$, $\oint_S \mathbf{A} \cdot d\mathbf{S} = \int_V \nabla \cdot \mathbf{A} dV$, where S is the closed surface bounding the volume V . The differential form also follows directly: apply the theorem to the small volume element of Fig. 4.3, whose surface integral is $q_{\text{enc}}/\epsilon_0$ by Gauss' law. Since the volume is arbitrary, the integrands must agree, which is Eq. (4.2).

4.2.1 Poisson's Equation

Applying ∇ twice gives the **Laplacian**

$$\nabla^2 \equiv \nabla \cdot \nabla = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$$

Combining the differential form of Gauss' law, Eq. (4.2), with the relationship between the electric potential and the electric field, Eq. (4.1), we immediately arrive at a second-order partial differential equation:

$$\nabla^2 V = \nabla \cdot \nabla V = -\frac{\rho}{\epsilon_0} \quad (4.3)$$

When $\rho = 0$ it reduces to Laplace's equation, $\nabla^2 V = 0$. This equation is of particular importance in practical applications (think: why?): in a conductor, or in any region bounded by conductors held at fixed potentials, the boundary conditions single out the physical solution, and Chapter 5 solves the equation by the method of image charges.

4.3 The Triangle

We have now assembled the three relations that tie together the electric field $\mathbf{E}$, the electric potential V , and the charge density ρ :

$$\begin{aligned}
\mathbf{E} &= -\nabla V & (\text{Eq. (4.1)}) \\
\nabla \cdot \mathbf{E} &= \frac{\rho}{\epsilon_0} & (\text{Eq. (4.2)}) \\
\nabla^2 V &= -\frac{\rho}{\epsilon_0} & (\text{Eq. (4.3)})
\end{aligned}$$

The first relation is purely geometric: the electric field is the downhill direction of the potential. The second is Gauss' law in differential form, and it is the one that knows about charge. The third follows from the other two once $\mathbf{E}$ is eliminated, and it is the one usually solved in practice, because it determines V directly from the charge distribution.

These three place $\mathbf{E}$, V and ρ at the vertices of a triangle, with one relation along each edge: the potential determines the field, the field determines the charge, and the charge determines the potential. Knowing any one of the three, together with the boundary conditions, fixes the other two. This is what makes electrostatics tractable — a problem that appears to require a vector field can often be reduced to a single scalar function V obeying one partial differential equation. The chapters that follow work around this triangle repeatedly.

4.4 Exercises

1. Prove equations (a) and (d) in Section 4.1.3, where f and g are scalar functions and $\mathbf{v}$ and $\mathbf{w}$ are vector functions, namely:

$$\nabla(fg) = (\nabla f)g + f(\nabla g)$$

and

$$\nabla \cdot (\mathbf{v} \times \mathbf{w}) = \mathbf{w} \cdot (\nabla \times \mathbf{v}) - \mathbf{v} \cdot (\nabla \times \mathbf{w})$$

2. An electric potential is given in Cartesian coordinates:

$$V = A(x^2 + y^2) + Bz^2,$$

where A and B are positive constants.

- Find the electric field $\mathbf{E}$.
 - Show that the electric field satisfies the fundamental property $\nabla \times \mathbf{E} = 0$.
 - Find the charge density $\rho(x, y, z)$ using the differential form of Gauss' law $\nabla \cdot \mathbf{E} = \rho/\epsilon_0$.
3. In Chapter 3 the electrostatic field was shown to satisfy $\oint \mathbf{E} \cdot d\mathbf{l} = 0$ around every closed path. This exercise connects that statement with $\nabla \times \mathbf{E} = 0$.

- A small rectangle lies in the plane $z = z_0$. It is centered at (x_0, y_0, z_0) , its sides Δx and Δy are parallel to the x and y axes, and it is traversed counterclockwise as seen from the $+z$ side. Following the method used in Sec. 4.2 for the flux through a small box, show that, to leading order in Δx and Δy ,

$$\oint \mathbf{E} \cdot d\mathbf{l} = \left(\frac{\partial E_y}{\partial x} - \frac{\partial E_x}{\partial y} \right) \Delta x \Delta y,$$

with the derivatives evaluated at the center.

- A field is proposed as $\mathbf{E} = E_0 (y/a) \hat{x}$ everywhere, where $E_0 > 0$ is a field strength and $a > 0$ a length. Find $\nabla \times \mathbf{E}$. Then calculate $\oint \mathbf{E} \cdot d\mathbf{l}$ directly around the square with corners $(0, 0, 0)$, $(L, 0, 0)$, $(L, L, 0)$ and $(0, L, 0)$, where $L > 0$ is a length, traversed in that order, and compare the result with (a). Can any arrangement of charges at rest produce this field?
4. A ball of radius R centered at the origin carries a total charge Q spread uniformly through its volume. Gauss' law gives its field as

$$\mathbf{E} = \frac{Q \mathbf{r}}{4\pi\epsilon_0 R^3} \quad (r < R), \quad \mathbf{E} = \frac{Q \mathbf{r}}{4\pi\epsilon_0 r^3} \quad (r > R),$$

where $\mathbf{r} = x\hat{x} + y\hat{y} + z\hat{z}$ and $r = |\mathbf{r}|$.

- Calculate $\nabla \cdot \mathbf{E}$ for $r < R$, and compare it with ρ/ϵ_0 .
- Calculate $\nabla \cdot \mathbf{E}$ for $r > R$, using identity (b) of Sec. 4.1.3 with $f = r^{-3}$ and $\mathbf{v} = \mathbf{r}$.
- The flux of $\mathbf{E}$ through a sphere of radius $b > R$ centered at the origin is Q/ϵ_0 , although $\nabla \cdot \mathbf{E} = 0$ at every point of that sphere and everywhere else outside the ball. Explain why this does not contradict Gauss' theorem of Sec. 4.2.

5. Take the electrostatic field, in a region that contains the box of part (b), to be

$$\mathbf{E}(x, y, z) = E_0 \left[\left(1 + \frac{xy}{a^2} \right) \hat{x} + \frac{x^2}{2a^2} \hat{y} \right],$$

where $E_0 > 0$ is a field strength and $a > 0$ a length.

- (a) Use $\nabla \cdot \mathbf{E} = \rho/\epsilon_0$ to find the charge density $\rho(x, y, z)$. Evaluate both ρ and $\mathbf{E}$ at the origin.
 - (b) Let S be the closed surface of the rectangular box $0 \leq x \leq a$, $0 \leq y \leq b$, $0 \leq z \leq c$, where $b > 0$ and $c > 0$ are lengths, with its normal pointing outward. Calculate the total electric flux Φ through S directly from its six faces.
 - (c) Calculate the enclosed charge q_{enc} by integrating ρ over the whole box, and verify Gauss' theorem of Sec. 4.2 and Gauss' law, $\Phi = q_{\text{enc}}/\epsilon_0$.
6. As an idealization, a volume charge distribution extends throughout space and is uniform in the y and z directions, with

$$\rho(x) = \rho_0 \frac{x}{a},$$

where $\rho_0 > 0$ and $a > 0$. Assume that the potential depends only on x . At the origin its value and the x component of the electric field are specified to be $V(0) = V_c$ and $E_x(0) = E_c$, where V_c is a real constant and $E_c < 0$.

- (a) Solve Poisson's equation for $V(x)$ and find the electric field $\mathbf{E}(x)$.
 - (b) Explain separately the physical roles of V_c and E_c . Show that the potential found in (a) gives, through Poisson's equation, the same $\rho(x)$ whatever the values of V_c and E_c .
 - (c) Find the two planes on which $\mathbf{E} = \mathbf{0}$, and evaluate the charge density on each plane.
7. Poisson's equation limits where a charged particle can rest in stable equilibrium.
- (a) Show that at any point of a region that contains no charge, the three second derivatives $\partial^2 V/\partial x^2$, $\partial^2 V/\partial y^2$ and $\partial^2 V/\partial z^2$ cannot all be positive.
 - (b) Four point charges, each of charge $Q > 0$, are fixed at $(a, 0, 0)$, $(-a, 0, 0)$, $(0, a, 0)$ and $(0, -a, 0)$, where $a > 0$. Show that $\nabla V = 0$ at the origin, and calculate the three second derivatives of V there.
 - (c) A small particle of charge $q > 0$ is placed at the origin. Is its equilibrium stable against a small displacement along x ? Along z ? Use (a) to show that, at a point where there is no other charge, no arrangement of fixed charges can make the second derivatives of a positive charge's potential energy positive along all three axes x , y and z .